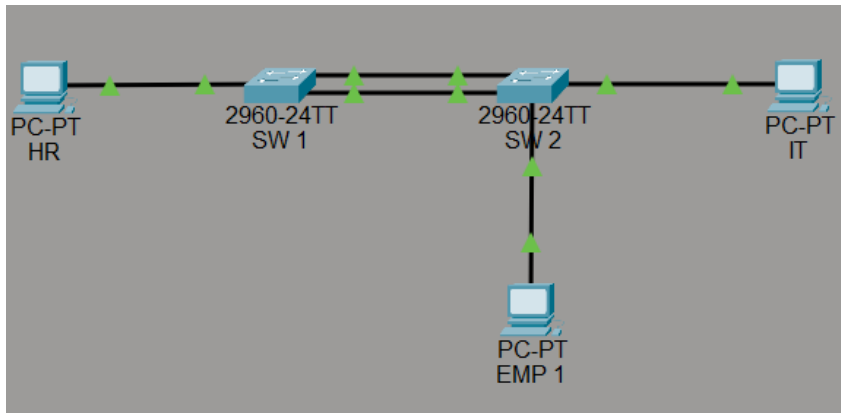


MINI PROJECT 5

EtherChannel using LACP + 802.1Q Trunking (Removing STP Blocking)

DIAGRAM:



DESCRIPTION:

This project explains how EtherChannel with LACP (active mode) bundles multiple physical links into one logical link to increase bandwidth and provide redundancy. It also shows how 802.1Q trunking works over the EtherChannel and how this setup prevents STP from blocking links by treating them as a single connection.

STEP 1: Create VLANs on SW 1,SW2:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs on SW1 & SW2:

SW1:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

SW 2:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 20
exit
interface fa0/2
switchport mode access
switchport access vlan 10
exit
```

STEP 3: Configure LACP EtherChannel

```
interface range fastEthernet0/23 - 24
channel-group 1 mode active
exit
```

STEP 4 : Configure Port-Channel as Trunk:

```
interface port-channel 1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

STEP 5: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1
PC-EMP 1	10	192.168.10.11/24	192.168.10.1

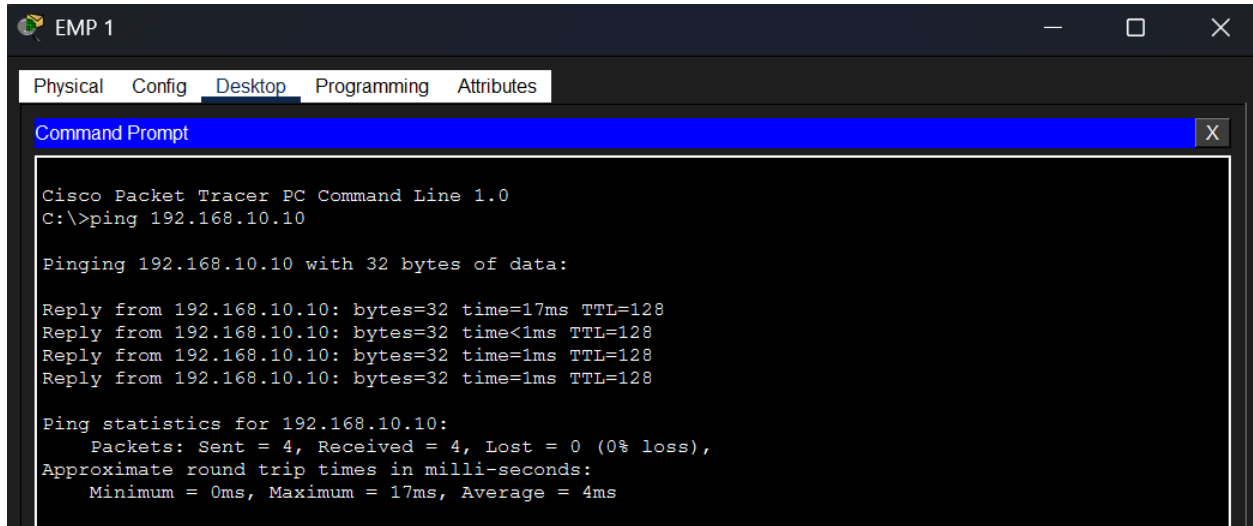
STEP 6: Verification & STP Observation:

show etherchannel summary

```
Group  Port-channel  Protocol  Ports
-----+-----+-----+-----
1      Po1 (SU)          LACP     Fa0/23 (P) Fa0/24 (P)
Switch#
```

show spanning-tree

STEP 7: Ping Test:



```
EMP 1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=17ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

we do not have Router or Layer 3 switch, so ping test from EMP-PC to HR.